UNIT 4 — 20 QUESTIONS

Motion Representation

Q61. The motion field of a scene is defined as:
- A. The set of all detected corners
- B. The projection of 3D scene velocities onto the 2D image plane
- C. The histogram of pixel intensities
- D. The camera's intrinsic matrix

Q62. Optical flow estimates:
- A. The true 3D motion field exactly
- B. The apparent motion of brightness patterns between frames
- C. The camera's focal length
- D. The colour distribution of a scene

Q63. The brightness constancy assumption states that:
- A. Pixel intensity changes linearly with depth
- B. A point's intensity remains constant as it moves between frames
- C. All pixels have equal brightness
- D. Brightness is inversely proportional to velocity

Q64. The "aperture problem" in optical flow arises because:
- A. The camera aperture is physically too small
- B. A single local gradient constraint gives one equation for two unknowns (u, v)
- C. Lighting is too bright
- D. The lens has excessive distortion

Q65. The Lucas-Kanade optical flow method assumes that:
- A. Flow varies significantly within a small neighbourhood
- B. Flow is constant within a small local neighbourhood
- C. Brightness constancy never holds
- D. Only global motion exists

Q66. The Horn-Schunck optical flow method extends the basic flow equation by adding a:
- A. Colour constancy term
- B. Global smoothness constraint
- C. Camera intrinsic term
- D. Morphological operator

Q67. Motion parallax explains why, from a moving viewpoint:
- A. Distant objects appear to move faster than near objects
- B. Near objects appear to move faster across the field of view than distant objects
- C. All objects appear stationary
- D. Colour shifts occur with distance

Q68. The general affine flow model for (u, v) has how many parameters?
- A. 2
- B. 4
- C. 6
- D. 8

Q69. Dense optical flow computes motion vectors:
- A. Only at a few selected feature points

B. At every pixel in the image
C. Only at image corners
D. Only along detected edges

Q70. Sparse optical flow (e.g., Lucas-Kanade on tracked points) computes motion vectors:
A. At every pixel
B. Only at a selected set of feature points
C. Only in the frequency domain
D. Only for the background

Q71. Differential optical flow techniques primarily rely on:
A. Image intensity derivatives (gradients)
B. Colour histograms
C. Camera calibration matrices
D. Morphological structuring elements

Q72. Feature-based motion estimation techniques track:
A. Every pixel uniformly
B. Distinct features such as corners or keypoints
C. Only the image's frequency spectrum
D. The camera's principal point

Q73. Regularization is added to optical flow estimation mainly to handle:
A. Textured regions with strong gradients
B. Textureless/ambiguous regions where flow is underdetermined
C. Colour correction
D. Camera calibration errors

Q74. Robust estimation techniques are especially important in optical flow when:
A. The scene is completely static
B. Multiple independently moving objects or outliers are present
C. The camera is perfectly calibrated
D. Only one object is visible

Q75. The brightness constancy assumption is most likely to fail under:
A. Constant, uniform lighting
B. Sudden illumination changes or specular reflections
C. Slow, smooth camera motion
D. High image resolution

Q76. The optical flow constraint equation is typically written as:
A. $Ix + Iy + It = 1$
B. $Ix \cdot u + Iy \cdot v + It = 0$
C. $u^2 + v^2 = It$
D. $Ix \cdot It = Iy \cdot u$

Q77. A pure rotational motion of a rigid object contributes to the motion field a component that is:
A. Always zero
B. Dependent on angular velocity and position
C. Independent of the object's position
D. Only visible in the frequency domain

Q78. In the Lucas-Kanade method, the local flow vector (u, v) is typically solved using:
A. A single linear equation

B. A least-squares solution over a local window
C. Fourier transformation
D. Morphological erosion

Q79. A completely textureless wall region in a video will typically produce optical flow estimates that are:
A. Highly accurate and unambiguous
B. Unreliable or ambiguous due to lack of gradient information
C. Always exactly zero
D. Only affected by colour, not motion

Q80. Which of the following is a feature-based (as opposed to purely differential) approach to motion estimation?
A. Horn-Schunck method
B. Tracking of matched keypoints/corners across frames
C. Direct pixel-wise brightness subtraction
D. Histogram equalization